

SUMMARY OF ARTICLES 1-2

Summary of the Main Ideas

1. Retrieval-Augmented Generation

“Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks”

By Patrick Lewis et al. (2020)

Original Paper:

[ArXiv Paper – Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks](#)

This paper introduced the original RAG architecture and became one of the foundational works behind modern AI assistants and retrieval-based LLM systems. ([arXiv](#))

Main Ideas of the Paper

A. Problem with Traditional Language Models

The authors explain that standard Large Language Models (LLMs):

- store knowledge only in model parameters,
- cannot easily update information,
- may hallucinate facts,
- and cannot explain where information came from.

Traditional models behave like “closed memory systems.”

B. Introduction of Retrieval-Augmented Generation (RAG)

The paper proposes combining:

1. **Parametric Memory**

- knowledge stored inside the neural network weights.

2. **Non-Parametric Memory**

- external document databases such as Wikipedia.

The RAG architecture retrieves relevant documents before generating responses. ([arXiv](#))

Core Architecture

Basic workflow:

User Query

↓

Retriever

↓

Relevant Documents

↓

Generator (LLM)

↓

Grounded Answer

The retriever searches a vector database using dense embeddings.

Document similarity commonly uses cosine similarity:

C. Two RAG Variants

1. RAG-Sequence

Uses the same retrieved documents for the entire generated response.

2. RAG-Token

Allows different retrieved documents for different generated tokens.

RAG-Token is more flexible but computationally expensive. ([arXiv](#))

D. Major Contributions

The paper demonstrated that RAG:

- improves factual accuracy,
- reduces hallucinations,
- generates more specific answers,

- improves explainability,
- and outperformed previous QA systems.

The authors achieved state-of-the-art performance in:

- open-domain question answering,
 - factual generation,
 - and knowledge-intensive NLP tasks. ([arXiv](#))
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E. Importance for Modern AI

This paper became foundational for:

- AI assistants,
- financial AI systems,
- enterprise search,
- legal AI,
- healthcare AI,
- and agentic AI systems.

Modern systems like:

- ChatGPT retrieval workflows,
 - financial copilots,
 - and enterprise AI search tools
all rely heavily on RAG concepts.
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2. Agentic AI

“OpenAI (2025). Agentic AI Systems and Autonomous Reasoning”

There does not appear to be a single formal paper with exactly this title publicly available. However, the topic is strongly represented across OpenAI’s 2025 releases and agentic AI research announcements. ([OpenAI](#))

The concepts are discussed in:

- OpenAI agent tool releases,
 - Operator,
 - autonomous AI systems,
 - and reasoning-based agents.
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Main Ideas of Agentic AI

A. Transition from Passive LLMs to Autonomous Agents

Traditional LLMs:

- answer prompts only.

Agentic AI systems:

- reason,
- plan,
- execute actions,
- use tools,
- maintain memory,
- and adapt dynamically.

This represents a shift from:

Prompt → Response

to:

Goal → Planning → Tool Use → Retrieval → Reasoning → Action

([IBM](#))

B. Multi-Step Reasoning

Agentic systems perform iterative reasoning.

Example workflow:

Goal

↓

Break task into subtasks

↓

Retrieve information

↓

Evaluate results

↓

Use tools/APIs

↓

Revise strategy

↓

Generate final output

This resembles human analytical thinking.

C. Tool Use and External Actions

Modern agents can:

- browse the web,
- query databases,
- execute code,
- use APIs,
- analyze spreadsheets,
- and interact with software systems.

OpenAI's Operator system is an example of an autonomous web agent. ([OpenAI](#))

D. Memory and Context Management

Agentic systems maintain:

- short-term memory,
- long-term memory,
- and contextual awareness.

Memory allows agents to:

- continue long tasks,
- remember goals,
- and adapt strategies.

E. Multi-Agent Collaboration

Modern architectures increasingly use:

- planner agents,
- retrieval agents,
- verification agents,
- and execution agents.

Each specialized agent handles part of a larger workflow. ([arXiv](#))

Importance for Investment Analysis

Agentic AI is highly relevant to finance because investment analysis requires:

- continuous information retrieval,
- reasoning under uncertainty,
- risk evaluation,
- forecasting,
- and explainable decisions.

An Agentic RAG system can:

1. retrieve SEC filings,
2. analyze market news,

- 3. forecast trends,
- 4. evaluate risks,
- 5. and generate investment recommendations autonomously.

Key Conceptual Difference

Traditional RAG	Agentic RAG
Single retrieval step	Iterative retrieval
Static prompting	Dynamic planning
One-pass response	Multi-step reasoning
Limited memory	Persistent memory
No autonomous tools	Tool execution
Simple QA	Autonomous analysis

Main Research Implications

These works together suggest the future of AI systems will involve:

- 1. Retrieval-based factual grounding
 - 2. Autonomous reasoning agents
 - 3. Multi-agent collaboration
 - 4. Financial decision intelligence
 - 5. Explainable AI systems
 - 6. Human-AI cooperative analysis
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Overall Conclusion

The Lewis et al. (2020) paper established the foundation for grounding LLMs using external knowledge retrieval. Agentic AI research extends this paradigm by adding autonomy, planning, reasoning, and tool execution.

Together, these concepts form the basis of modern:

- AI copilots,
- financial AI analysts,
- autonomous investment systems,
- and next-generation decision-support platforms. ([arXiv](#))